Arithmetic Progressions

The big idea

An **arithmetic progression (AP)** is a list of numbers where each term is obtained by adding the same fixed number, the **common difference d**, to the previous term. Saving ₹50 more each month, seats in successive rows, taxi fares per kilometre: all APs.

AP IN ONE LINE

$a, a + d, a + 2d, a + 3d, \ldots$ where a is the first term and $d = a_2 - a_1 = a_3 - a_2 = \ldots$

The nth term

GENERAL TERM

$$a_n = a + (n - 1)d$$

WHICH TERM OF THE AP 21, 18, 15, … IS −81?

$a = 21, d = -3$. $21 + (n - 1)(-3) = -81 \Rightarrow (n - 1) = 34 \Rightarrow \mathbf{n = 35}$.

The nth term **from the end** of an AP with last term l is $l - (n - 1)d$.

Sum of the first n terms

SUM OF N TERMS

$$S_n = n/2\,[2a + (n - 1)d] = n/2\,(a + l), \text{ where } l \text{ is the last term}$$

The story behind it: young Gauss added $1 + 2 + \ldots + 100$ by pairing the first and last terms $(1 + 100)$, second and second-last $(2 + 99)$… there are 50 pairs of 101, so the sum is 5050.

FIND THE SUM OF THE FIRST 22 TERMS OF 8, 3, −2, …

$a = 8, d = -5, n = 22$. $S_{22} = 11[16 + 21 \times (-5)] = 11 \times (-89) =$ **−979**.

USEFUL FACTS

The sum of the first n natural numbers is $n(n + 1)/2$.

$a_n = S_n - S_{n-1}$, which lets you find a term when you know the sum formula.

MIXING UP N AND A_n

n is the **position** of a term (always a positive whole number); a_n is the **value** of that term.

Word problem

A SUM OF ₹700 IS TO BE USED FOR SEVEN PRIZES, EACH ₹20 LESS THAN THE PREVIOUS. FIND EACH PRIZE.

$S_7 = 700, d = -20$: $7/2\,[2a + 6(-20)] = 700 \Rightarrow 2a - 120 = 200 \Rightarrow a = 160$.

Prizes: **₹160, 140, 120, 100, 80, 60, 40**.

Quick recap

- d is the constant difference between consecutive terms.
- $a_n = a + (n - 1)d$.
- $S_n = n/2\,[2a + (n - 1)d] = n/2\,(a + l)$.

Practice questions

1. The common difference of the AP 3, 1, −1, −3, … is
   (a) 2 (b) −2 (c) 4 (d) −4

2. The 11th term of the AP −3, −1/2, 2, … is
   (a) 19 (b) 22 (c) 28 (d) 25

3. How many two-digit numbers are divisible by 3?
   (a) 29 (b) 30 (c) 31 (d) 33

4. The sum $1 + 2 + 3 + \dots + 50$ equals
   (a) 1250 (b) 1275 (c) 1300 (d) 2550

5. If $S_n = 3n^2 + 2n$ for an AP, its first term is
   (a) 3 (b) 5 (c) 8 (d) 2

Answers: 1. b 2. b 3. b 4. b 5. b